

Automated face mask detection is critical for public health. This project implements the CL-SSDXcept pipeline, combining a Caffe SSD face detector with a fine-tuned Xception classifier. Using CLAHE preprocessing and transfer learning on the RFMD dataset, our model achieves 99.1% accuracy, running in real-time and significantly outperforming baseline architectures like VGG16 and ResNet50.

Face Mask Detection in Real-Time: A CL-SSDXcept Approach

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01. Introduction

- Problem: Traditional monitoring methods struggle with lighting variations and pose changes.
- Objective: Implement a robust, real-time pipeline to classify mask usage from live webcam feeds.
- Why Xception? Depthwise separable convolutions provide high accuracy with low computational cost, making it ideal for real-time inference.

02. Objective

To build a highly accurate, real-time face mask detection pipeline (CL-SSDXcept) capable of processing live webcam feeds. By combining Caffe SSD for face localization, CLAHE preprocessing to normalize harsh lighting, and a fine-tuned Xception deep learning model, the system aims to reliably classify mask usage (Mask vs. No Mask) without latency to automate public health monitoring.

03. Methodology

- Stage 1 (Detection): Pre-trained Caffe SSD extracts facial bounding boxes from webcam frames.
- Stage 2 (Preprocessing): CLAHE (Contrast-Limited Adaptive Histogram Equalization) is applied to the LAB color space (L-channel) to normalize lighting without distorting mask colors. Faces are resized to 299×299.
- Stage 3 (Classification): A fine-tuned Xception model (top 20 layers unfrozen, trained via GPU) classifies the ROI as "Mask" or "No Mask."

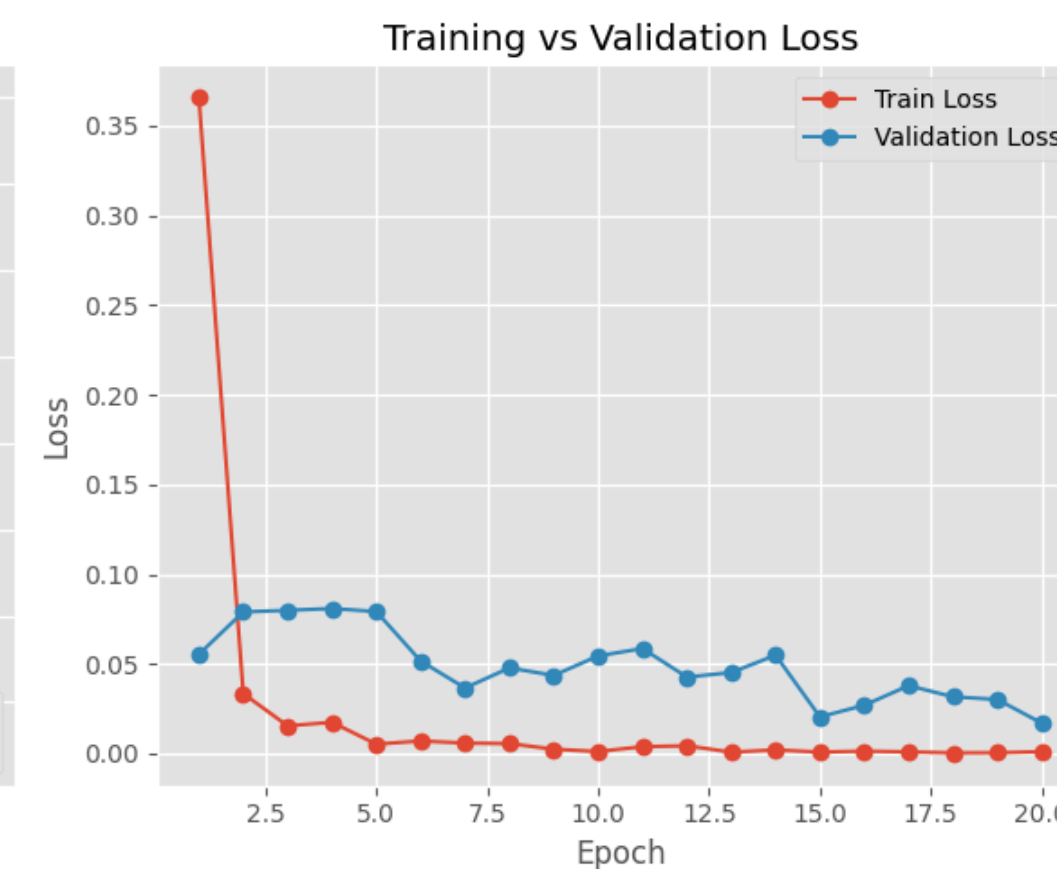
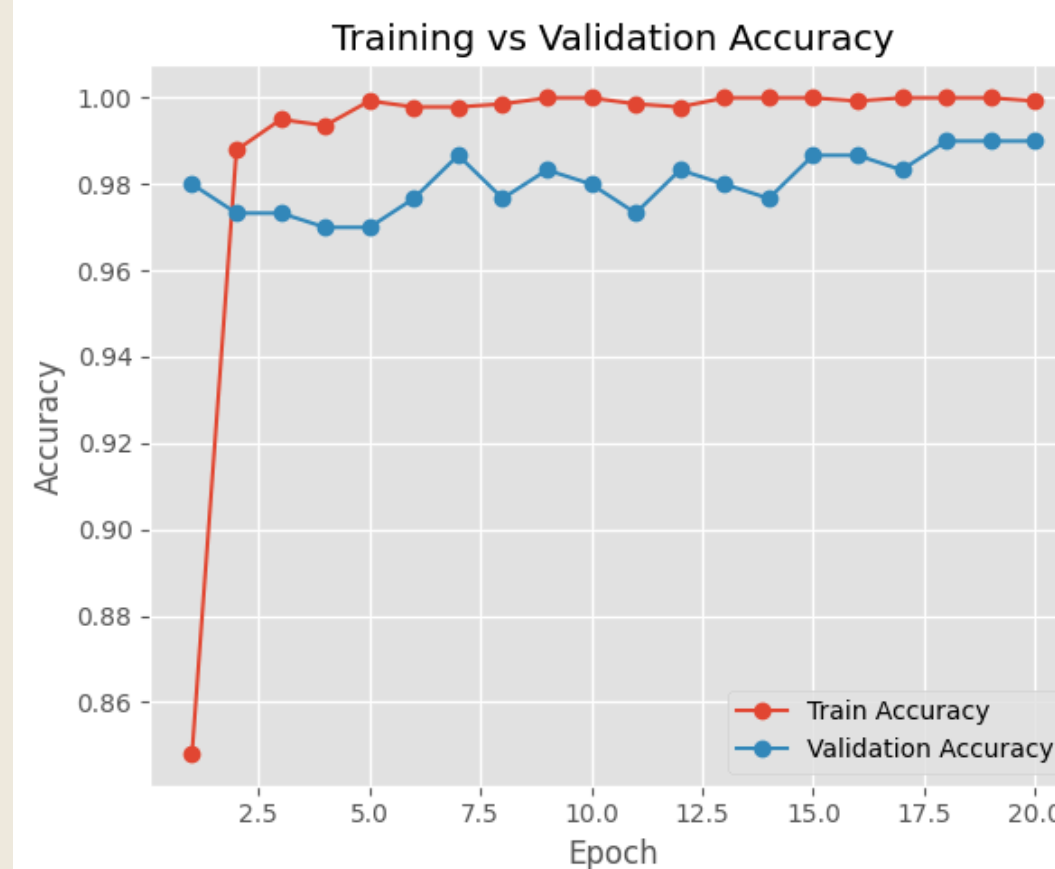
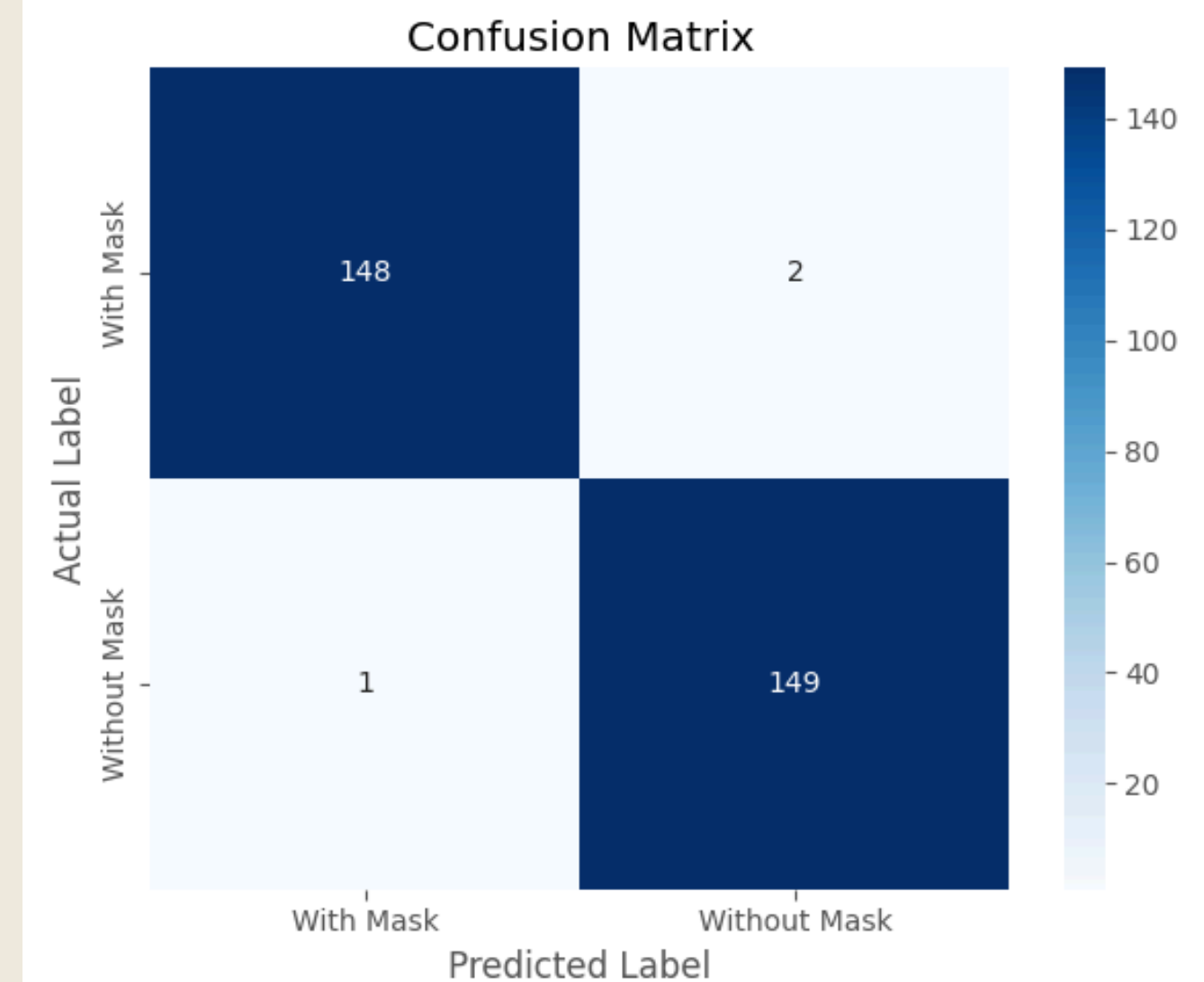
04. Results/Findings

Model Comparison (Validation Accuracy):

- Xception (Ours): 99.10% (Optimal)
- MobileNetV2: 72.67% (Overfitting)
- VGG16: 53.67% (Poor Convergence)
- ResNet50: 50.00% (Random Guessing)

Demo, Video

https://drive.google.com/file/d/1OVwzonb-2AyvPgXRbQCqKg8UuMB5UeOo/view?usp=drive_link



05. Conclusion

- Architectural Superiority: Xception is vastly superior at isolating facial boundaries and mask textures compared to standard convolutions (VGG16) or deep residual blocks (ResNet50).
- Fine-Tuning is Critical: A frozen baseline model achieved only 66% accuracy. Unfreezing the top 20 layers and utilizing GPU acceleration was required to achieve the final 99.1% accuracy.